



1. Find the general antiderivative of the following functions.

1).  $f(x) = \frac{1}{2} + \frac{3}{4}x^2 - \frac{4}{5}x^3$

2).  $f(x) = 6\sqrt{x} - \sqrt[6]{x}$

3).  $f(x) = \sqrt[4]{x^3} + \sqrt[3]{x^4}$

4).  $f(x) = \frac{2}{\sqrt{1-x^2}}$

2. Find the following indefinite integrals.

1).  $\int (3e^x + 7 \sec^2 x) dx$

2).  $\int (\cos \theta - 5 \sin \theta) d\theta$

3).  $\int \frac{5 - 4x^3 + 2x^6}{x^6} dx$

4).  $\int \frac{2 + x^2}{1 + x^2} dx$

**3.** Given  $f''(x) = 4 \sin x + 3 \cos x$ ,  $f(0) = 3$ ,  $f'(0) = 4$ , find  $f(x)$ .